Fixed-G Cluster Inference in Bester, Conley, and Hansen

Monte Carlo Figures for a Spatial Moving-Average Design

Graduate Econometrics Simulation Note

May 12, 2026

0.1 Overview

These figures illustrate Bester, Conley, and Hansen's fixed-G argument in the spatial moving-average Monte Carlo. The key point is not that cluster scores converge to constants. Within each large group, the score obeys a central limit theorem; across groups, the scores become asymptotically independent. With G fixed, the cluster covariance estimator (CCE) is therefore built from only G approximately independent random scores, so its limit remains random.

The simulation uses the paper's spatial moving-average design on a $K \times K$ lattice:

$$x_s = \sum_{\|h\| \leq 2} \gamma^{\|h\|} v_{s+h}, \qquad \varepsilon_s = \sum_{\|h\| \leq 2} \gamma^{\|h\|} u_{s+h},$$

with $y_s = x_s \beta + \varepsilon_s$ and $\beta = 1$. The null hypothesis is true in every Monte Carlo replication.

0.2 Figure A: CCE t-statistic

The null is true, so a conventional plug-in argument would suggest a $N(0,1)$ reference distribution. BCH instead keep the number of groups fixed and let each group grow. The raw CCE statistic has limiting reference distribution

$$\sqrt{\frac{G}{G-1}}\, t_{G-1}.$$

For $G = 4$, this is much wider than $N(0,1)$, because the denominator is estimated from only four approximately independent group scores.

0.3 Figure B: Rejection Rates

The dashed line is the nominal 5% rejection rate. Procedures using $N(0,1)$ critical values treat the variance estimate as essentially known. The BCH procedure rescales the CCE t-statistic by $\sqrt{(G-1)/G}$ and uses t_{G-1} critical values, which accounts for fixed-G variance-estimation uncertainty.

0.4 Figure C: Variance Estimator

The plotted object is

$$\left(Q^{-1} \hat{V}_N Q^{-1}\right)_{\text{slope,slope}},$$

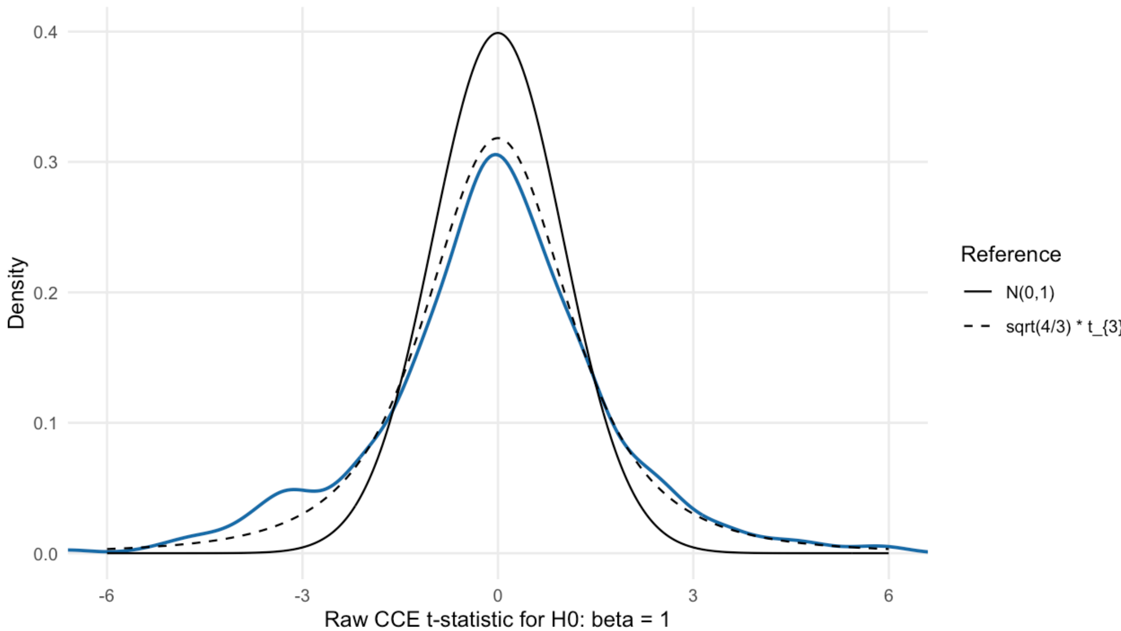

Figure 1: Raw CCE t-statistic under the null. The blue density is the Monte Carlo distribution for K = 36, gamma = 0.6, and G = 4. The dashed reference is the BCH fixed-G limit.

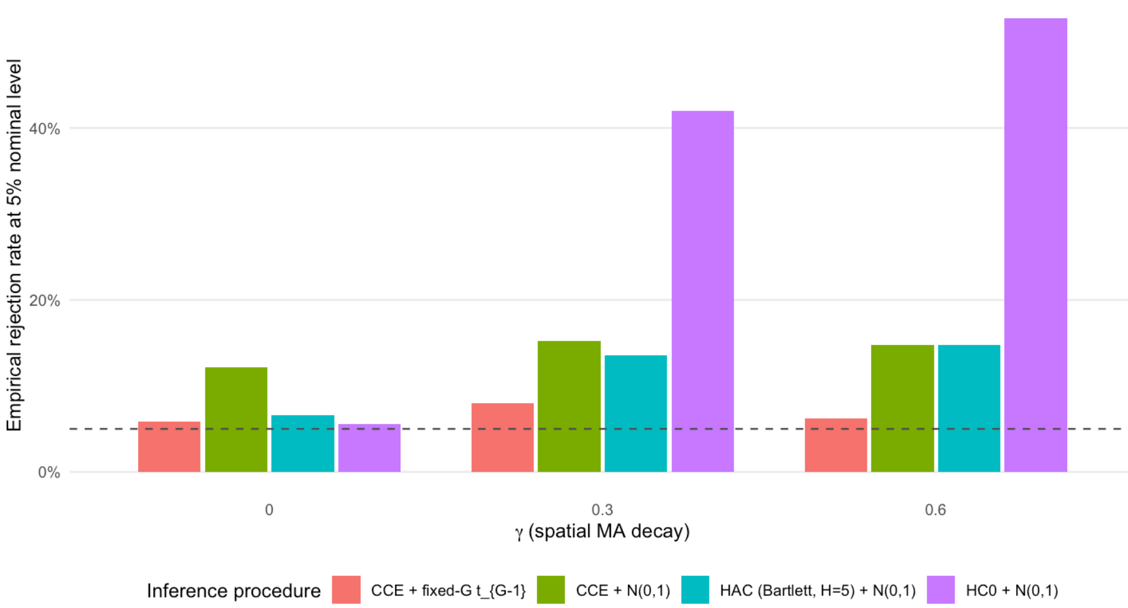

Figure 2: Empirical rejection rates for nominal 5 percent tests under the true null. The dashed horizontal line is the target size.

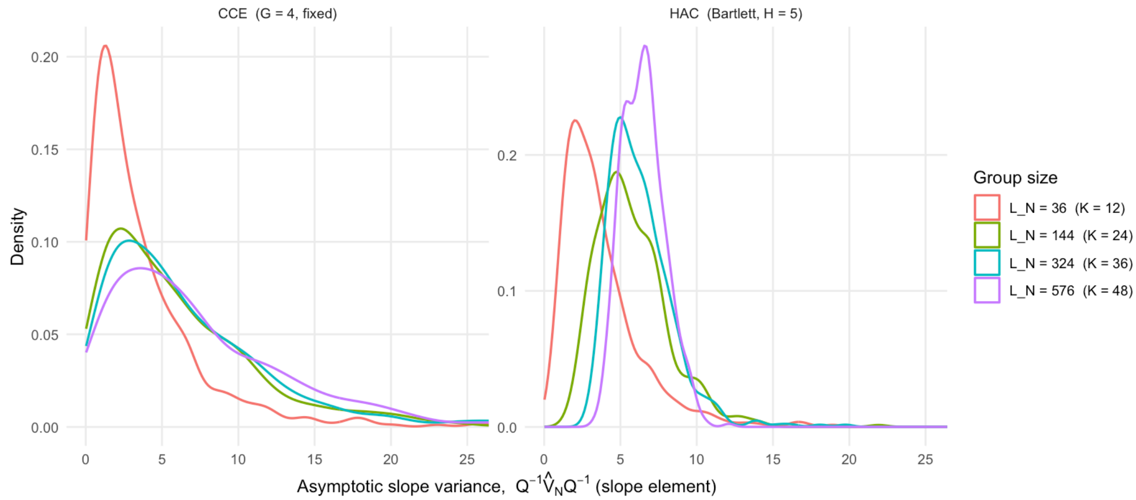

Figure 3: Sampling distribution of the asymptotic-scale slope variance estimate. The CCE panel keeps G fixed while group size grows.

the asymptotic variance scale for $\sqrt{N}(\hat{\beta} - \beta)$. This is deliberately not divided by N. The actual variance of $\hat{\beta}$ shrinks like $1/N$ for any sensible estimator, so plotting that object would miss the BCH point.

The CCE density remains dispersed as L_N grows because fixed G leaves only G pieces of information about the group-score variance. The HAC panel is a local-covariance contrast whose sampling variation concentrates as the lattice grows. In this simulation H is fixed at 5, so it should be read as a concentration comparison rather than an exact implementation of a growing-bandwidth HAC sequence.

0.5 Takeaway

The figures separate two ideas that are easy to conflate. First, large groups justify approximately normal and independent group scores. Second, fixed G means the variance estimator is still random, because it is based on only G group-level pieces of information. BCH's fixed-G critical values target that second source of uncertainty.